

Knowledge and Data

Working Group

Week 1: Formal Systems and Logic



Question 1—Notation

Consider the following formula, which is in *infix* notation:

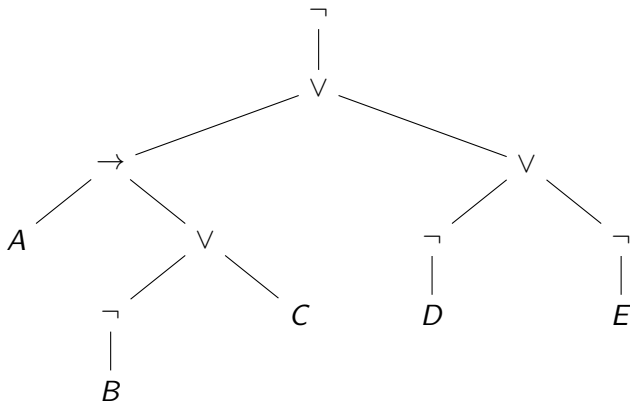
$$\neg((A \rightarrow (\neg B \vee C)) \vee (\neg D \vee \neg E))$$

Which of the following is an equivalent representation in *prefix* notation?

1. $\neg(\vee(\rightarrow(E, \vee(\neg(B), D)), \vee(\neg(C), \neg(A))))$
2. $\vee(\neg(\vee(A, \rightarrow(\neg(B), C)), \vee(\neg(D), \neg(E))))$
3. $(((((E)\neg, (D)\neg)\vee, ((C, (B)\neg)\vee,) \rightarrow)\vee)\neg$
4. $\neg(\vee(\rightarrow(\vee(\vee(\vee(\neg(\neg(\neg(A, B, C, D, E))))))))$
5. $\neg(\vee(\rightarrow(A, \vee(\neg(B), C)), \vee(\neg(D), \neg(E))))$

Question 1—Notation

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4. $\neg(\vee(\rightarrow(\vee(\vee(\vee(\neg(\neg(\neg(A, B, C, D, E))))))))$
5. $\neg(\vee(\rightarrow(\mathbf{A}, \vee(\neg(\mathbf{B}), \mathbf{C})), \vee(\neg(\mathbf{D}), \neg(\mathbf{E}))))$

Question 2—Entailment in Arithmetic

It is easy to see that formulae $f : 3x < 12 + 6y$ and $g : y = 1$ imply (entail) formula $h : x < 10$. That is, $f, g \models h$.

Give the formal argument why this is the case according to the model theoretic semantics given during the lecture.

Do NOT solve this using calculus!

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From the lecture:

- Truth is defined in terms of an assignment for variables.
- Let V be the set of variables, then $I^V : V \rightarrow \mathbb{N}$ is an assignment; a function that assigns natural numbers to each variable in V .
- Function I^V is a model of a formula f if $I^V(f)$ is true.
- Formula f entails g , $f \models g$, iff g is true for all models of f .

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Give the formal argument why this is the case according to the model theoretic semantics given during the lecture.

- $x < 10$ is entailed by the other two formulae if h is true for all the models of f and g
- These models are assignments of variables x and y for which f and g are true
- That is, $I^v(3x < 12 + 6y, y = 1)$ and $I^v(x < 10)$ are true for all possible assignments
- If $I^v(3x < 12 + 6y, y = 1)$ is true, then also $I^v(3x < 18)$ is true, or $I^v(x < 6)$
- Therefore, also $x < 10$

Question 3—Entailment in Propositional Logic

In Propositional Logics, which of the following entailments is true:

a) $\neg p \vee (q \rightarrow p) \models \neg p \wedge q$

b) $\neg p \vee (q \rightarrow p) \models \neg p \vee q$

1. **a** and **b** are true
2. **a** is true; **b** is false
3. **a** is false; **b** is true
4. **a** and **b** are false

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Our truth table:

p	q	$\neg p$	$q \rightarrow p$	$\neg p \vee (q \rightarrow p)$	$\neg p \wedge q$	$\neg p \vee q$
F	F	-	-	-	-	-
F	T	-	-	-	-	-
T	F	-	-	-	-	-
T	T	-	-	-	-	-

for all assignments that make the left hand side true, the right side must also be true

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Question 4—Models in Arithmetic

Definition

A model of a knowledge base KB is an assignment that satisfies all its axioms/formulae

Example: a formula $x < 6$ is satisfied if $x = 3$, which makes $x = 3$ a model of KB . However, $x = 8$ does **not satisfy** formula $x < 6$, and is thus **not a model** of KB .

Question 4—Models in Arithmetic

Which of the following assignments I are models for these two formulae:

a) $2x + (4 + y) - z = 10$

b) $x + y < 7$

1. $I(x) = 3, I(y) = 4, I(z) = 4$

2. $I(x) = 3, I(y) = 3, I(z) = 4$

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Question 5—Formal Systems: Syntax

Task: design a very simple new Formal System, L :

- The only thing you can say in this language is whoops and wow
e.g. whoops, wow, wow, whoops, whoops, whoops, ...
- Use inductive definition to define its syntax
- Define the following three elements:
 1. the basic propositions P
 2. the base case, i.e. the most simple sentence(s)
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 $\phi, \psi \in L \rightarrow [\phi, \psi] \in L$

Question 6—Formal Systems: Semantics

Consider, once again, our formal system L

Your friends know that if you type more whoops than wow in your messages, then you are **unhappy**, whereas otherwise you are **happy**.

Task: come up with a definition of the semantics of L

- for any well-formed sentence, you must be able to determine whether you are happy or unhappy
- Answer the following:
 1. Let $p \in P$; what would be a sensible interpretation of $I(p)$?
 2. Let s be a sentence in L ; what is the interpretation of $I(s)$?

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 $I(\text{wow}) = 1$, $I(\text{whoops}) = -1$
NB: the choice of $[-1, 1]$ is arbitrary here
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 2. Let s be a sentence in L ; what is the interpretation of $I(s)$?
 $I([\text{whoops}, \text{tail}]) = -1 + I(\text{tail})$
 $I([\text{wow}, \text{tail}]) = 1 + I(\text{tail})$
here *tail* denotes the remainder of s

Question 7—Facts and Reproduction

Which of the following sentences is true (according to Forbes)?

1. The most time consuming problem for data scientists is to find the right algorithm for pattern recognition.
2. Without sufficient data, tacit knowledge is useless in practices.
3. There is usually more time spend on preparing, linking and cleaning data than on building the datasets in the first place.
4. Knowledge graphs are the only way to make tacit knowledge explicit

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Quiz

To test your proficiency:

Exercise quiz Module 1
(see the course's Canvas page)

NB: completing the quiz is mandatory to continue to next week's module